27th National High School Physics Finals: Solutions

I. (25 points) Fill in the blanks

1. $\dfrac{4\pi^2 kR}{4\pi^2 k - M\omega^2}$ （6 points）

2. $\dfrac{fV}{V+u}$, $\dfrac{fV}{V-u}$ （4 points）

3. $T_0\left(1-\dfrac{Mg}{p_0 A}\right)^{\frac{1-1}{\gamma}}$ （6 points）

4. $7.3\times10^6\,\mathrm{m\cdot s^{-1}}$, $2.8\times10^8\,\mathrm{m\cdot s^{-1}}$, $8.4\times10^5\,\mathrm{V}$ （9 points）

$$p=\frac{h}{\lambda}$$

$$E=\sqrt{p^2c^2+m_e^2y}$$

$$E_k=eV=E-m_ec^2$$

II. (20 points)

1. Let m be the block mass. Its acceleration down the incline is

$$a=\frac{mg\sin\theta-\mu_1 mg\cos\theta}{m}=g\left(\sin\theta-\mu_1\cos\theta\right),\qquad(1)$$

The speed of the block at the bottom of the incline is

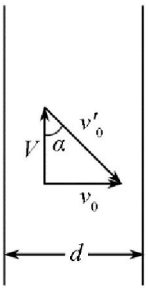
$$v_0=\sqrt{\frac{2ah}{\sin\theta}}=\sqrt{2gh\left(1-\mu_1\cot\theta\right)}=4.0\,\mathrm{m/s}\qquad(2)$$

In the belt frame, the initial speed of the block entering the belt is

$$v_0'=\sqrt{v_0^2+V^2}=5.0\,\mathrm{m/s}\text{。}\qquad(3)$$

The angle α between the motion and the belt edge satisfies

$$\tan\alpha=\frac{4}{3}\text{。}\qquad(4)$$



The block decelerates on the belt with acceleration magnitude

$$a'=\frac{\mu_2 mg}{m}=\mu_2 g\text{。}\qquad(5)$$

The distance traveled relative to the belt before coming to rest is

$$s'=\frac{v_0'^2}{2a}=\frac{v_0'^2}{2\mu_1 g},\qquad(6)$$

The least friction coefficient μ_2 preventing escape over the belt edge satisfies

$$s'\sin\alpha = \frac{v_0'^2 \sin\alpha}{2\mu_2 g} = d \qquad (7)$$

Hence

$$\mu_2 = \frac{v_0'^2 \sin\alpha}{2gd} = 0.5\text{。} \qquad (8)$$

2. The block's friction force on the belt has magnitude

$$F = \mu_2 \cdot \frac{\eta g}{\mu_2 g} v_0' = \eta v_0', \qquad (9)$$

It points along v_0'. In the ground frame, belt speed is V. The block's work on the belt per unit time is

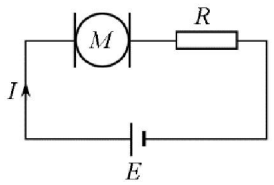
$$W = -FV\cos\alpha, \qquad (10)$$

Thus the additional power required by the load is

$$\Delta P = -W = \eta V^2 = 640\text{W}\text{。} \qquad (11)$$

The motor's no-load output power is

$$P_0 = I_0 E - I_0^2 R = 360\text{W}\text{。} \qquad (12)$$



Its output power with cargo is

$$P = P_0 + \Delta P = 1000\text{W}\text{。} \qquad (13)$$

Let I be the operating current with cargo. Then

$$P = IE - I^2 R, \qquad (14)$$

Solving gives

$$I = 10\text{A}\text{。} \qquad (15)$$

Scoring (8) (12) (13) (14) (15) total ; ; ; ;

2 (15) (18) (10) ; ; .

III. (20 points)

The light rod has length $2l$. At angle θ to the horizontal, ball 1, have speeds v_1, v_2. The rod's angular speed is ω. Conservation of mechanical energy gives

$$0 = mgl\sin\theta - 2mgl\sin\theta + \frac{1}{2}(2m)v_1^2 + \frac{1}{2}mv_2^2\text{。} \qquad (1)$$

Also,

$$v_1 = v_2 = l\omega, \qquad (2)$$

Solving (1)(2) gives

$$\omega = \sqrt{\frac{2g\sin\theta}{3l}} \qquad (3)$$

The angular momentum of the rod and two balls about the pivot is

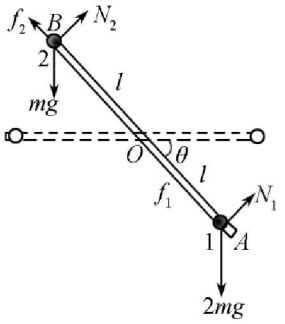
$$L = 2mlv_1 + mlv_2 , \qquad (4)$$

The angular-momentum law gives

$$2mgl\cos\theta - mgl\cos\theta = \frac{\Delta L}{\Delta t} 。 \qquad (5)$$

By the definition of angular acceleration β,

$$\beta = \frac{\Delta\omega}{\Delta t} , \qquad (6)$$



Equations (2)(4)(5)(6) give

$$\beta = \frac{g\cos\theta}{3l} 。 \qquad (7)$$

While both are on the rod, they circle the pivot. Ball 1 has the following tangential and normal accelerations:

$$a_{1t} = l\beta \qquad (8)$$

$$a_{1t} = l\omega \qquad (9)$$

Let N_1 be the force normal to the rod on ball 1, and f_1 the force on ball 1 along the rod. Newton's second law gives

$$2mg\cos\theta - N_1 = 2ma_{1t} , \qquad (10)$$

$$f_1 - 2mg\sin\theta = 2ma_{1t} \qquad (11)$$

Equations (3)(9)(10)(11) give

$$N_1 = \frac{4}{3}mg\cos\theta 。 \qquad (12)$$

$$f_1 = \frac{10}{3}mg\sin\theta 。 \qquad (13)$$

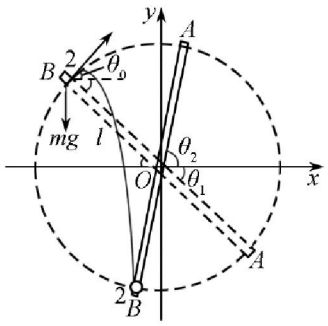
For ball 2, the force normal to the rod on ball 2 has magnitude N_2; the force along the rod on ball 2 has magnitude f_2. The same analysis gives

$$N_2 = \frac{4}{3}mg\cos\theta , \qquad (14)$$

$$f_2 = \frac{1}{3}mg\sin\theta 。 \qquad (15)$$

2 By (13)(14)(15), balls 1 and 2 have equal static-friction limits; show that ball 1 needs more friction than ball 2. During rotation, ball 1 reaches the limit first, so ball 1 slides first. As ball 1 starts to slide, let the angle

of the rod to the horizontal be θ_1. Then $f_1(\theta_1)=\mu N_1(\theta_1)$,



that is, $\frac{10}{3}mg\sin\theta_1=\frac{3}{4}\mu mg\cos\theta_1$,

(16)

Substituting the data into (16) gives

$\theta_1=\frac{\pi}{6}$。

(17)

At $\theta=\theta_1$, ball 1 starts sliding outward. Ball 2 starts next to the end, so ball 2 leaves after negligible time. Thus the rod's angle when ball 1 leaves is also $\theta_1=\frac{\pi}{6}$。

When ball 1 leaves, the massless rod exerts no force on ball 2. Ball 2 then feels gravity alone. By (2007), its initial speed in projectile motion is

$$v_0=l\sqrt{\frac{2g\sin\theta_1}{3l}}=\sqrt{\frac{3gl}{3}}。\qquad(18)$$

The initial velocity makes the following angle with the horizontal:

$$\theta_0=\frac{\pi}{2}-\theta_1=\frac{\pi}{3}。\qquad(19)$$

Ball 2 stays on the rod during flight; neither exerts a force. The rod follows without affecting it. Ball 2 leaves when its distance from the pivot is again l. Take the axes Oxy shown. At time t, taking the start of ball 2's flight as time zero, ball 2 has coordinates

$$x=-l\cos\theta_1+v_0\cos\theta_0 t,\qquad(20)$$

$$y=l\sin\theta_1+v_0\sin\theta_0 t-\frac{1}{2}gt^2,\qquad(21)$$

When ball 2 leaves the rod,

$$l^2=x^2+y^2。\qquad(22)$$

Using (17)(18)(19) gives

$$t^2\left(t^2-2\sqrt{\frac{l}{g}}t-\frac{2}{3}\frac{l}{g}\right)=0,\qquad(23)$$

Solving gives

$$t=\left(1+\frac{\sqrt{15}}{3}\right)\sqrt{\frac{l}{g}}。\qquad(24)$$

At this time,

$$\begin{cases} x = -\dfrac{2\sqrt{3}-\sqrt{5}}{6} l \\ y = -\dfrac{2+\sqrt{15}}{6} l \end{cases}。 \qquad (25)$$

Let θ_2 be the rod's angle to the horizontal when ball 2 leaves (see figure). Then

$$\cos\theta_2 = \frac{|x|}{l} = \frac{2\sqrt{3}-\sqrt{5}}{6}, \qquad (26)$$

$$\theta_2 = \arccos\left(\frac{2\sqrt{3}-\sqrt{5}}{6}\right) = 78.2° \text{ (or } 1.36 \text{ rad)}. \qquad (27)$$

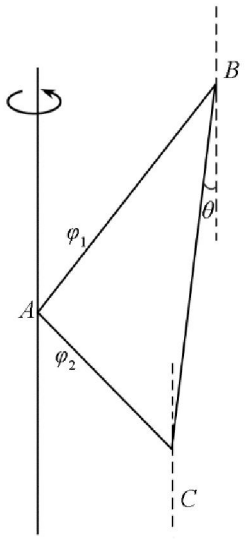
Scoring: (26) (15) ; ; each ; ; ;

(20) (27) ; ; each ; each .

IV. (15 points)

Method 1:

Let m be B's mass, θ the angle of BC to the vertical, and ω its angular speed. For pair BC, repulsion is f_{BC}. Newton's law gives



$$f_{AB}\sin\varphi_1 - f_{BC}\sin\theta = m\omega^2|AB|\sin\varphi_1, \qquad (1)$$

$$f_{AB}\cos\varphi_1 - f_{BC}\cos\theta = 0, \qquad (2)$$

$$f_{AC}\sin\varphi_2 + f_{BC}\sin\theta = m\omega^2|AC|\sin\varphi_2, \qquad (3)$$

$$f_{AC}\cos\varphi_2 - f_{BC}\cos\theta = 0。 \qquad (4)$$

(2)、(4) Using in (1)、(3) gives

$$\frac{f_{AB}\sin(\varphi_1-\theta)}{f_{AC}\sin(\varphi_2+\theta)} = \frac{|AB|\sin\varphi_1}{|AC|\sin\varphi_2}。 \qquad (5)$$

The geometry gives

$$\frac{|AB|}{AC} = \frac{\sin(\varphi_2+\theta)}{\sin(\varphi_1-\theta)} \qquad (6)$$

With the given expressions for f_{AB} and f_{BC}, (5) gives

$$\left(\frac{|AB|}{|AC|}\right)^{a-2} = \frac{\sin\varphi_1}{\sin\varphi_2} \qquad (7)$$

Also, (2)、(4) give $\dfrac{f_{AB}}{f_{AC}} = \dfrac{\cos\varphi_2}{\cos\varphi_1}$。 (8)

Substituting the given expressions for f_{AB} and f_{BC} gives

$$\frac{|AB|^{\alpha}}{|AC|^{\alpha}} = \frac{\cos\varphi_2}{\cos\varphi_1}。 \qquad (9)$$

Combining （7)、(9) gives

$$\sin^{\alpha}\varphi_1\cos^{\alpha-2}\varphi_1=\sin^{\alpha}\varphi_2\cos^{\alpha-2}\varphi_2\text{。}\qquad(10)$$

If $\varphi_1\neq\varphi_2$, the following equation must have two distinct roots

$$\sin^{\alpha}\varphi\cos^{\alpha-2}\varphi-C=0\qquad(11)$$

in $\left(0,\frac{\pi}{2}\right)$, for a suitable constant C. The function $\sin^{\alpha}\varphi\cos^{\alpha-2}\varphi$ must therefore be nonmonotonic on $\left(0,\frac{\pi}{2}\right)$; $\sin^{\alpha}\varphi$ and $\cos^{\alpha-2}\varphi$ cannot both be increasing or both decreasing. As φ grows, if $\sin^{\alpha}\varphi$ rises, $\cos^{\alpha-2}\varphi$ must fall; if $\sin^{\alpha}\varphi$ falls, $\cos^{\alpha-2}\varphi$ must rise. Thus α and $\alpha-2$ have the same sign, giving

$$\alpha<0\qquad(12)$$

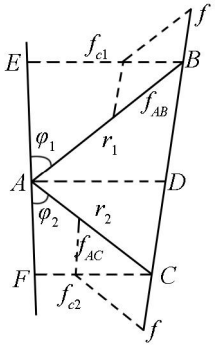
$$a>2\text{。}\qquad(13)$$

For $a<0$, $\sin^{-\alpha}\varphi\cos^{2-\alpha}\varphi$ is zero at $\varphi=0$ and $\frac{\pi}{2}$. Hence $\sin^{-\alpha}\cos^{2-\alpha}\varphi$ on $\left(0,\frac{\pi}{2}\right)$ has an extremum there. Thus （11) has at least two distinct roots for a suitable C. Likewise for $a>2$. Satisfying either of （12)、(13) is necessary and sufficient.

Scoring (62)(10)(1) ~ (4) each ; ; ; and the argument below them: 4 points in total.

Method 2:

In the figure, B、C repel with f; attractions from A are f_{AB}、f_{AC}. Their centripetal forces are f_{C1}、f_{C2}; distances to A are r_1、r_2. Similar triangles give



$$\frac{f_{AB}}{r_1}=\frac{f_{C1}}{AB}=\frac{f}{BD},\qquad(1a)$$

$$\frac{f_{AC}}{r_2}=\frac{f_{C2}}{AD}=\frac{f}{CD}\text{。}\qquad(2a)$$

Dividing the two equations gives

$$\frac{f_{AB}}{f_{AC}}\frac{r_2}{r_1}=\frac{f_{C1}}{f_{C2}}=\frac{CD}{BD}\qquad(3a)$$

The given conditions imply

$$\frac{f_{AB}}{f_{AC}}=\left(\frac{r_1}{r_2}\right)^{\alpha},\qquad(4a)$$

$$\frac{f_{C1}}{f_{C2}}=\frac{EB}{FC}=\frac{r_1\sin\varphi_1}{r_2\sin\varphi_2},\qquad(5a)$$

$$\frac{CD}{BD}=\frac{AF}{AE}=\frac{r_2\cos\varphi_2}{r_1\cos\varphi_1},\qquad (6a)$$

Substituting (3a)~(6a) into gives

$$\left(\frac{r_1}{r_2}\right)^{\alpha-1}=\frac{r_1\sin\varphi_1}{r_2\sin\varphi_2}=\frac{r_2\cos\varphi_2}{r_1\cos\varphi_1}。\qquad (7a)$$

From (7a),

$$\sin^{\alpha}\varphi_1\cos^{\alpha-2}\varphi_1=\sin^{\alpha}\varphi_2\cos^{\alpha-2}\varphi_2。\qquad (8a)$$

The remaining argument is the same as in "Method 1".

Scoring: see "Method 1".

V. (15 points)

Ambient pressure is p_a. Initially the air has pressure p_a, temperature T_a, volume V_a. Contact with the iceberg brings equilibrium at temperature T_1 and volume V_0. This is an isobaric process, so

$$\frac{V_0}{T_1}=\frac{V_a}{T_a}\qquad (1)$$

The change in the air's internal energy during this process is

$$\Delta U=2.5p_a\left(V_0-V_a\right),\qquad (2)$$

The work done by the atmosphere on the air is

$$W=-p_a\left(V_0-v_a\right)\qquad (3)$$

Let Q be the heat from the iceberg to the air. The first law of thermodynamics gives

$$Q=\Delta U-W。\qquad (4)$$

The four preceding equations give

$$Q=3.5p_aV_a\left(\frac{T_1-T_a}{T_a}\right)\qquad (5)$$

In (5), Q is negative: heat actually flows from the air to the iceberg.

After cooling to the ice temperature, the air is heated at fixed volume V_0, from T_1 to ambient temperature T_a, absorbing heat from its surroundings. If pressure is p_1,

$$\frac{p_1}{T_a}=\frac{p_a}{T_1}\qquad (6)$$

Let u be the nozzle volume. The first jet lowers vessel pressure from p_1 to p_2. The given conditions imply

$$p_1\left(V_0-u\right)=p_2V_0,\qquad (7)$$

so $p_2 = p_1 \dfrac{V_0 - u}{V_0}$ （8）

The kinetic energy gained by the expelled gas is

$\Delta E_{k1} = (p_1 - p_a)u$。 （9）

After the second jet, vessel pressure falls from p_2 to p_3. The given conditions yield

$p_3 = p_2 \dfrac{V_0 - u}{V_0}$ （10）

The kinetic energy gained by the expelled gas is

$\Delta E_{k2} = (p_2 - p_a)u$。 （11）

After jet N, vessel pressure falls from p_N to p_{N-1}. The given conditions yield

$p_{N+1} = p_N \dfrac{V_0 - u}{V_0}$ （12）

The kinetic energy gained by the expelled gas is

$\Delta E_{1k} = (p_N - p_a)u$。 （13）

If vessel pressure reaches ambient pressure after N jets, then

$p_{N+1} = p_a$， （14）

Jetting now stops. The total kinetic energy of all the expelled gas is

$E_k = \Delta E_{k1} + \Delta E_{k2} + \cdots + \Delta E_{kN}$ （15）

Using (13) to (15), becomes

$E_k = p_1 u \left[1 + \left(\dfrac{V_0 - u}{V_0} \right) + \left(\dfrac{V_0 - u}{V_0} \right)^2 + \cdots + \left(\dfrac{V_0 - u}{V_0} \right)^{N-1} \right] - N p_a u$， （16）

The bracket in term on the right of （16） is an N-term geometric series, so

$E_k = p_1 u \dfrac{1 - \left(\dfrac{V_0 - u}{V_0} \right)^N}{1 - \dfrac{V_0 - u}{V_0}} - N p_a u$。 （17）

Also, （8）、（10）、（12）、（14） give

$p_1 \left(\dfrac{V_0 - u}{V_0} \right)^N = p_a$， （18）

Taking natural logarithms of both sides of （18） gives

$N \ln \left(1 - \dfrac{u}{V_0} \right) = \ln \dfrac{p_a}{p_1}$。 （19）

Since $u \ll V_0$, the approximation $\ln(1 + x) \approx x$ further simplifies （19） to

$$N=\frac{V_0}{u}\ln\frac{p_1}{p_a} \tag{20}$$

Equations (17)、(18)、(20) give

$$E_{\mathrm{k}}=(p_1-p_a)V_0-p_aV_0\ln\frac{p_1}{p_a} \tag{21}$$

Substituting (121)(6) into gives

$$E_{\mathrm{k}}=p_av_a\left(1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}\right)。 \tag{22}$$

The electrical energy obtainable from this kinetic energy is

$$E=0.45p_aV_a\left(1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}\right)。 \tag{23}$$

Obtaining E requires the iceberg to absorb $-Q$ of heat. Melting it absorbs a total of

$$Q_1=mL。 \tag{24}$$

Hence the total electrical energy obtainable is

$$E_1=\frac{mL}{-Q}E。 \tag{25}$$

Substituting (5)(22)and into gives

$$E_1=\frac{9}{70}mL\frac{1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}}{1-\frac{T_1}{T_a}}, \tag{26}$$

Substituting the numerical data gives

$$E_1=1.5\times10^{14}\,\mathrm{J} \tag{27}$$

(22)(28)(7)(9)(3) (5) ; ; ; ; ;

3; (25) ~ (27) each .

Method 2:

At ambient pressure p_a, let n moles of air undergo the process as all the ice melts.

In step (a), volume changes by ΔV and temperature by $\Delta T=T_1-T_a$. The air releases the heat

$$Q=-p_a\Delta V-\frac{5}{2}p_a\Delta V。 \tag{1}$$

On the right, term one is atmospheric work on the system; term two is the internal energy change. The equation of state gives

$$Q=\frac{7}{2}nR\left(T_a-T_1\right),\tag{2}$$

This heat equals the latent heat absorbed as the iceberg melts, so

$$Q=mL,\tag{3}$$

Combining (2)、(3) gives

$$n=\frac{2mL}{7R\left(T_a-T_1\right)}。\tag{4}$$

In isochoric heating step (b), let the final pressure be p_0 and volume V_0. Then

$$p_a=\frac{T_a p_a}{T_1},\tag{5}$$

$$V_0=\frac{nRT_a}{p_0}\tag{6}$$

In each small isothermal jet step, the vessel pressure changes by Δp, with

$$\frac{p+\Delta p}{p}=\frac{V_0-u}{V_0},\tag{7}$$

Here V_0 is the volume in step (b), and p the vessel pressure. The expelled gas has kinetic energy

$$\Delta E=\left(p-p_a\right)u,\tag{8}$$

Eliminating u using (7) gives

$$\Delta E_k=-\left(p-p_a\right)V_0\frac{\Delta p}{p}。\tag{9}$$

Taking $\Delta E\to dE$, $\Delta p\to dp$, the total kinetic energy is

$$E_{\mathrm{k}}=\int_{p_0}^{p_a}-\left(p-p_a\right)V_0\frac{dp}{p}\tag{10}$$

$$=\left(p_0-p_a\right)V_0-p_aV_0\ln\frac{p_0}{p_a}$$

The total electrical energy obtained under the given conditions is

$$E=0.45E_{\mathrm{k}},\tag{11}$$

Substituting (11)(5)、(6)、(10) into gives

$$E=\frac{9}{70}mL\frac{1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}}{1-\frac{T_1}{T_a}};\tag{12}$$

Substituting the numerical data gives

$$E=1.5\times10^{14}\ \mathrm{J}。\tag{13}$$

Scoring: use the criteria for Method 1.

1. Before its first collision with a plate, the ball has kinetic energy and speed

$$E_{k1} = 0 \quad (1)$$

$$v_1 = 0 \quad (2)$$

Just after this collision, the charge on the ball is

$$q = C_0 U \quad (3)$$

Its kinetic energy and speed are then

$$E'_{k1} = 0 \quad (4)$$

$$v'_1 = 0 \quad (5)$$

The electric force accelerates it toward the other plate. Just before its second collision, its kinetic energy is

$$E_{k2} = E'_{k2} + qU \quad (6)$$

Using (3)、(4) gives

$$E_{k2} = C_0 U^2 \quad (7)$$

Its speed just before the second collision is

$$v_2 = \sqrt{\frac{2E_k}{m}} \quad (8)$$

Its speed and kinetic energy just after the second collision are

$$v'_2 = ev_2 \quad (9)$$

$$E'_{k2} = \frac{1}{2} m v'^2_2 \quad (10)$$

From (9)、(10),

$$E'_{k2} = e^2 E_{k2} \quad (11)$$

Its kinetic energy just before the third collision with a plate is

$$E_{k2} = E'_{k2} + qU \quad (12)$$

From (3)、(7)(11)、(12),

$$E_{k3} = \left(1 + e^2\right) C_0 U^2 \quad (15)$$

Its speed just before the third collision with a plate is

$$v_1 = \sqrt{\frac{2E_{k2}}{m}} \quad (16)$$

Its speed and kinetic energy just after the third collision are

$$v_3' = ev_3 \tag{17}$$

$$E_{k3}' = e^3 E_{k3} \tag{18}$$

Before its fourth collision with a plate, the ball has kinetic energy

$$E_{k4} = E_{k3}' + qU \tag{19}$$

From (3)、(15)、(18)、(19),

$$E_{k4} = \left(1 + e^2 + e^4\right) C_0 U^2 \tag{20}$$

Its speed just before the fourth collision is

$$v_4 = \sqrt{\frac{2E_{k4}}{m}} \tag{21}$$

Its speed and kinetic energy just after the fourth collision are

$$v_4' = ev_4 \tag{22}$$

$$E_{k4}' = e^2 E_{k4} \tag{23}$$

Continuing likewise, its kinetic energies just before and after collision n are

$$E_{kn} = \left[1 + e^2 + \cdots + e^{2(n-2)}\right] C_0 U^2 \tag{24}$$

$$E_{kn}' = e^2 \left[1 + e^2 \cdots + e^{2(n-2)}\right] C_0 U^{22} \tag{25}$$

Thus

$$E_{kn} = \frac{1 - e^{2(n-1)}}{1 - e^2} C_0 U^2 \tag{26}$$

$$E_{ln}' = \frac{e^2\left[1 - e^{2(n-1)}\right]}{1 - e^2} C_0 U^2 \tag{27}$$

For inelastic collisions, $e < 1$; E_{kn} and E_{kn}' grow with the collision count. As $n \to \infty$,

$$E_{k?} = \frac{1}{1 - e^2} C_0 U^2 \tag{28}$$

$$E_{k\infty}' = \frac{e^2}{1 - e^2} C_0 U^2 \tag{29}$$

The maximum kinetic energy is the limiting value just before a collision, so

$$E_{kmax} = E_{k\infty} = \frac{1}{1 - e^2} C_0 U^2 \tag{30}$$

2. After a sufficiently long time, as $n \to \infty$, the speeds just after one collision and just before the next are

$$v_\infty' = \sqrt{\frac{2E_{k\infty}'}{m}} = eU\sqrt{\frac{2C_0}{\left(1 - e^2\right)m}} \tag{31}$$

$$v_\infty = \sqrt{\frac{2E_{k\infty}}{m}} = U\sqrt{\frac{2C_0}{(1-e^2)m}} \tag{32}$$

The time interval between these collisions is

$$T = \frac{d}{\frac{v'_\infty + v_\infty}{2}} \tag{33}$$

Hence the average current through the external circuit is

$$I = \frac{q}{T} \tag{34}$$

From (31)、(32)、(33)、(34),

$$I = \frac{C_0 U^2}{d}\sqrt{\frac{1+eC_0}{1-e2m}} \tag{35}$$

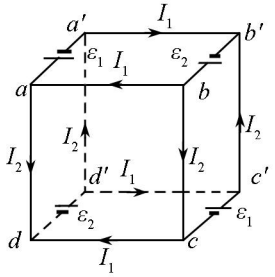
Points: 8 (2×3+2×3+2) or : total; : ; : each;
2 (35): .

VII.

Method 1:

1. The induced emfs in $a'a$ and cc' are

$$E_{a'a} = E_{cc'} = \varepsilon_1 = \frac{\sqrt{2}}{2}Bl^2\omega\sin\theta \tag{1}$$



The induced emfs in $b'b$ and dd' are

$$\varepsilon_{b'b} = \varepsilon_{dd'} = \varepsilon_2 = \frac{\sqrt{2}}{2}Bl^2\omega\cos\theta \tag{2}$$

Circuit symmetry gives

$$I_{ba} = I_{a'b'} = I_{d'c'} = I_{cd} = I_{1'},\quad I_{ad} = I_{d'a'} = I_{c'b'} = I_{bc} = I_2 \tag{3}$$

Kirchhoff's current law gives

$$I_{aa'} = I_{c'c} = I_1 - I_2 \tag{4}$$

$$I_{b'b} = I_{dd'} = I_1 + I_2 \tag{5}$$

Kirchhoff's voltage law gives

$$I_1 R + I_{aa'}R + I_1 R_1 + I_{bb'}R = \varepsilon_2 - \varepsilon_1 \tag{6}$$

$$I_1 R + I_{dd'}R + I_2 R - I_{ad}R = \varepsilon_2 + \varepsilon_1 \tag{7}$$

$$I_{bc} = I_{ad} = I_1 = \frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}(\cos\theta - \sin\theta) \tag{8}$$

$$I_{ad} = I_{bc} = I_2 = \frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}(\cos\theta + \sin\theta) \tag{9}$$

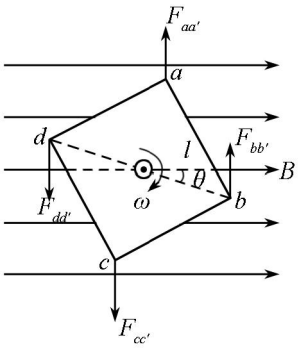
2. At any orientation, the angle of diagonal db to the field is θ. In $a'a$、cc', and $b'b$、dd', the currents are

$$I_{a'a}=I_{ad}-I_{ba}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\sin\theta \qquad (10)$$

$$I_{cc'}=I_{bc}-I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{4}\sin\theta \qquad (11)$$

$$I_{b'b}=I_{ba}+I_{bc}=\frac{\sqrt{2}}{4}\frac{BI^2\omega}{R}\cos\theta \qquad (12)$$

$$I_{dd'}=I_{ad}+I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (13)$$



The torque needed for uniform rotation is the net magnetic torque on the circuit, so

$$F_{a'a}=F_{cc'}=BlI_{a'a},\quad F_{b'b}=F_{dd'}=BlI_{b'b} \qquad (14)$$

$$M=2F_{a'a}\frac{\sqrt{2}}{2}l\sin\theta+2F_{b'b}\frac{\sqrt{2}}{2}l\cos\theta \qquad (15)$$

Substituting (11)~(14) into gives

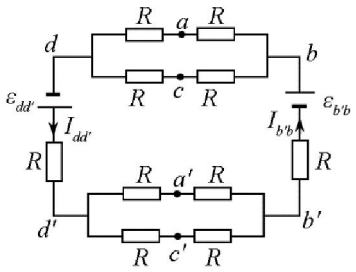
$$M=\frac{B^2l^4\omega}{2R}。 \qquad (16)$$

(40) (6) (4) (5)、(2)： ； ： ； ： ； ：

1 (2) (5) (4) ： ； ： ； ： .

Method 2:

1. The induced emfs in $a'a$ and cc' are



$$\varepsilon_{a'a}=\varepsilon_{cc'}=\frac{\sqrt{2}}{2}Bl^2\omega\sin\theta \qquad (1b)$$

The induced emfs in $b'b$ and dd' are

$$\varepsilon_{b'b}=\varepsilon_{dd'}=\frac{\sqrt{2}}{2}Bl^2\omega\cos\theta \qquad (2b)$$

With $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ shorted, only $\varepsilon_{b'b}$, $\varepsilon_{dd'}$ act. If $a'a$, cc' are opened, the circuit is as shown. The currents in $b'b$ and dd' are

$$I_{b'b}^{(1)}=I_{dd'}^{(1)}=\frac{\varepsilon_{b'b}+\varepsilon_{dd'}}{4R}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (3b)$$

The currents in ba、ad、bc and cd are

$$I_{bd}^{(1)}=I_{ad}^{(1)}=I_{cd}^{(1)}=\frac{1}{2}I_{b'b}^{(1)}=\frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}\cos\theta \qquad (4b)$$

By symmetry, the potential differences across a'、a and across c、c' are

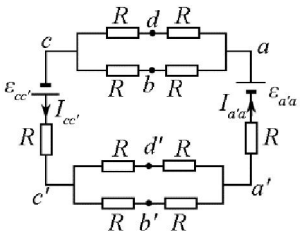
$$U_{a'a} = U_{cc'} = 0 \qquad (5b)$$

After connecting $a'a$ and cc', the currents in $a'a$ and cc' are

$$I_{aa'}^{(1)} = I_{cc'}^{(1)} = 0 \qquad (6b)$$

Thus connecting $a'a$, cc' leaves the currents in ba, ad unchanged.

Now only $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ act; $\varepsilon_{b'b}$, $\varepsilon dd'$ are shorted. Find each branch current. Opening $b'b$, dd' gives the circuit shown. As above, with $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ alone, currents in $a'a$, cc' are



$$I_{a'a}^{(2)} = I_{cc'}^{(2)} = \frac{\sqrt{2}}{4}\frac{Bl\omega^2}{R}\sin\theta \qquad (7b)$$

The currents in ba、 ad、 bc and cd are

$$I_{ba}^{(2)} = I_{cd}^{(2)} = -\frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\sin\theta \qquad (8b)$$

$$I_{ad}^{(2)} = I_{bc}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\sin\theta \qquad (9b)$$

Across b'、 b and d、 d', $U_{b'b} = U_{dd'} = 0$. On reconnecting $b'b$, dd', currents in $b'b$ and dd' are

$$I_{b'b}^{(2)} = I_{dd'}^{(2)} = 0 \qquad (10b)$$

Thus connecting bb', dd' leaves all branch currents unchanged.

By superposition, the currents in ba、 ad、 bc and cd are

$$I_{ba} = I_{ba}^{(1)} + I_{ba}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}(\cos\theta - \sin\theta) \qquad (11b)$$

$$I_{ad} = I_{ad}^{(1)} + I_{ad}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}(\cos\theta + \sin\theta) \qquad (12b)$$

$$I_{bc} = I_{bc}^{(1)} + I_{bc}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}(\cos\theta + \sin\theta) \qquad (13b)$$

$$I_{cd} = I_{cd}^{(1)} + I_{cd}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}(\cos\theta - \sin\theta) \qquad (14b)$$

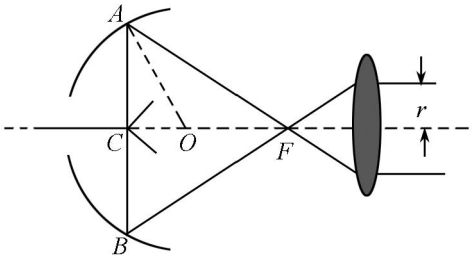
2. As in Method 1.

Scoring: use the criteria for Method 1.

VIII.

Solution:

1. Below, rays reflected by the spherical mirror meet at the convex lens focus.

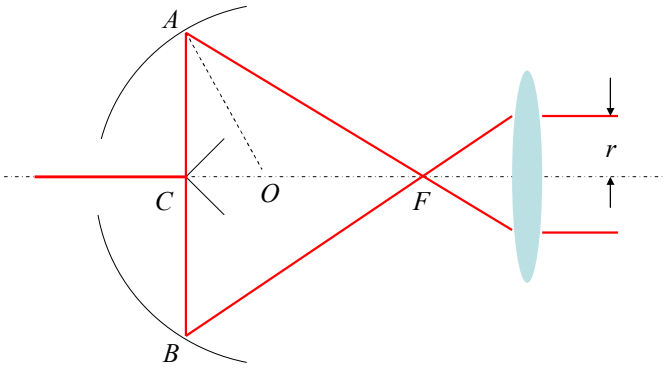


2. The ray diagram gives $CO = x$. Let $\angle CAO = a$; then geometry gives:

$$\sin\alpha = \frac{x}{R}, \qquad (1)$$

Solution:

1、 Below, rays reflected by the spherical mirror meet at the convex lens focus.



2、 The ray diagram gives $CO = x$. Let $\angle CAO = \alpha$; geometry gives:

$$\sin\alpha = \frac{x}{R}, \tag{1}$$

$$r = f\cot(2\alpha) \quad . \tag{2}$$

Combining the two equations gives

$$x = R\sin(\frac{1}{2}\arctan\frac{f}{r}). \tag{3}$$

Grading: Correct ray diagram: 5 points; equation （3）: 5 points.